

Bryan Zeng

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EXPERIENCE

INTEGRITY Security Services

Boston, MA

Full-Stack Software Engineer

January 2024 – December 2025

- Architected AWS-based full-stack web applications (Node.js, React) that support the certificate provisioning and lifecycle management for 100K+ IEEE 1609.2 certificates
- Engineered cross-platform mobile apps (Kotlin, Swift) enabling real-time V2X interoperability between roadside units, vehicles, and mobile devices for construction zone hazard detection and worker safety alerts
- Built cryptographic signing libraries (Kotlin, Swift) implementing ECDH key exchange for secure V2X message authentication, enabling tamper-proof communication across connected transportation infrastructure
- Optimized system observability by migrating 32M+ logs to DocumentDB, accelerating queries and cutting overhead
- Containerized EV2G app and ensured reliable CI/CD by creating Jenkins pipelines for build, deployment, and testing
- Developed Flask microservice for traffic intersection validation, enabling Python-based validation from the web app
- Consolidated three repos into a monorepo to facilitate end-to-end testing and increase reusability and extensibility

Wolters Kluwer - UpToDate

Waltham, MA

Product Full-Stack Software Engineer

January 2023 – December 2023

- Led modernization initiatives (Java, Spring Boot, Vue.js) eliminating technical debt and resolving critical security vulnerabilities for a medical resources platform serving 3M+ health professionals
- Enhanced internal developer tooling for 150+ developers to ensure proper backend configuration, reducing setup errors and accelerating onboarding
- Redesigned error-handling front-end architecture across application, improving user experience and maintainability

NURover

Boston, MA

Software Developer | Webmaster

September 2021 – May 2025

- Developed haptic control interface in C++ as proof-of-concept for teleoperated 4-DOF robotic arm manipulation
- Implemented Python ROS state machine for autonomous navigation with Zed2 camera and IMU/GPS sensor fusion

PROJECTS

Large-Scale Misinformation Classification

September 2025 – Present

- Trained deep learning models (CNN, LSTM, FNN, Longformer) for misinformation detection across 3.4M articles, achieving 99.6% accuracy with Longformer outperforming baseline SVM/XGBoost by 12%

Cheffy: AI Cooking Assistant

September 2025 – Present

- Co-designed AI agentic search cooking assistant using Gemini and multi-agent RAG to autonomously retrieve recipes and provide personalized cooking guidance

Finance Tracker AI Chatbot

June 2025 – Present

- Created RAG-powered chatbot using LangChain for automated financial analysis transaction categorization and personalized budget insights

Twitter Sentiment Analysis Stock Portfolio

October 2022 – December 2022

- Constructed quantitative trading model using NLP sentiment analysis (NLTK, Scikit-Learn) on **1.75M+ tweets**

EDUCATION

Northeastern University - Khoury College of Computer Sciences

Boston, MA

M.S. in Computer Science | GPA: 4.0/4.0

Expected May 2026

B.S. in Computer Science and Business Administration | GPA: 3.966/4.0

Key Coursework: Artificial Intelligence, Machine Learning, Software Engineering, ML Operations, Cybersecurity, Networks, Database Management, Systems, Algorithms & Data, Object-Oriented Design, Data Science

TECHNICAL SKILLS

- **Languages:** JavaScript/TypeScript, Java, Python, Kotlin, Swift, Go, SQL, C++
- **Frameworks:** React, Node.js, Vue, Spring Boot, Flask
- **AI/ML Tools:** PyTorch, TensorFlow, Scikit-Learn, LangChain, Ollama
- **Developer Tools:** AWS (CloudFormation, EC2, S3, Lambda), Docker, Jenkins, Git, Jira, MongoDB